

2(a)	<table><tr><th>description of phenotypic variation</th><th>cause of variation</th></tr><tr><td>Tomato plants grown in a glasshouse and grown outside vary in the yield of tomatoes they produce. Seventeen genes associated with tomato yield have been identified.</td><td>$V_G + V_E$</td></tr><tr><td>New strawberry plants from the variety called Sweet Ann are made by asexual reproduction. The new plants grow to different sizes and produce different numbers of fruit.</td><td>V_E</td></tr><tr><td>The domestic cat has a blood group system with three possible blood types: A, B and AB. The blood types are determined by antigens present on the cell surface membrane of red blood cells.</td><td>V_G</td></tr><tr><td>Over 50 genes have variants that are associated with excessive weight-gain in humans. Other risk factors for excessive weight-gain include diet and exercise.</td><td>$V_G + V_E$</td></tr><tr><td>Resting heart rate in humans varies between different individuals. Some factors that influence resting heart rate include: biological sex, family history of heart disease, number of cigarettes smoked, medication taken.</td><td>$V_G + V_E$</td></tr></table> five correct = 3 marks ;;; four or three correct = 2 marks two or one correct = 1 mark	description of phenotypic variation	cause of variation	Tomato plants grown in a glasshouse and grown outside vary in the yield of tomatoes they produce. Seventeen genes associated with tomato yield have been identified.	$V_G + V_E$	New strawberry plants from the variety called Sweet Ann are made by asexual reproduction. The new plants grow to different sizes and produce different numbers of fruit.	V_E	The domestic cat has a blood group system with three possible blood types: A, B and AB. The blood types are determined by antigens present on the cell surface membrane of red blood cells.	V_G	Over 50 genes have variants that are associated with excessive weight-gain in humans. Other risk factors for excessive weight-gain include diet and exercise.	$V_G + V_E$	Resting heart rate in humans varies between different individuals. Some factors that influence resting heart rate include: biological sex, family history of heart disease, number of cigarettes smoked, medication taken.	$V_G + V_E$	3
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2(b)	mutation ;	1												

Question	Answer	Marks
2(c)	any three from: 1 crossing over ; 2 independent / random, assortment ; 3 random, fusion of gametes / fertilisation ; 4 random mating ;	3